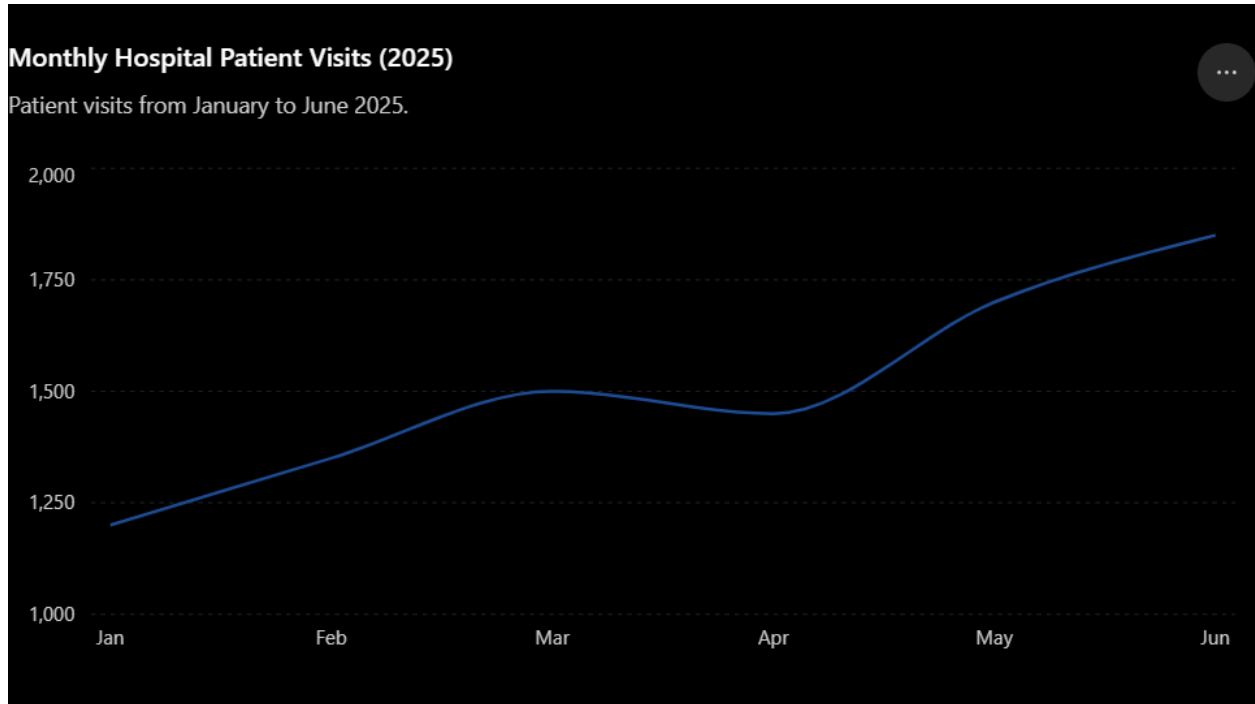


## Class Activity

### 1. Case Study: Monthly Hospital Patient Visits



1. Which visualization is most suitable?

Answer: Line Chart, because it shows the trend of patient visits over time (month-wise).

2. What trend can be observed?

Answer: The number of patient visits shows an overall increasing trend from January to June, with a slight decrease in April.

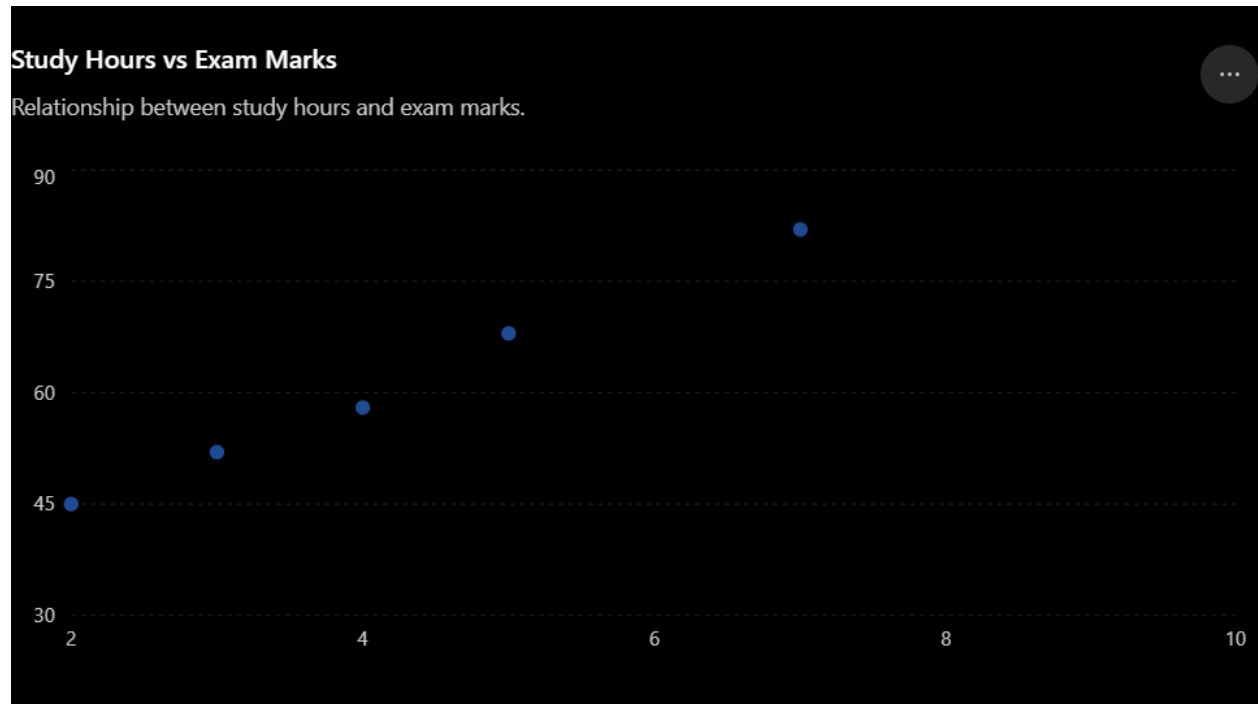
3. Identify the month with the highest patient count.

Answer: June has the highest patient count with 1850 patients.

4. Predict whether visits are increasing or decreasing.

Answer: The patient visits are increasing overall, indicating a rising trend in hospital visits.

### 2. Case Study: Study Hours vs Exam Marks



1. Which chart is appropriate?

Answer: Scatter Plot, because it shows the relationship between study hours and exam marks.

2. Is there a positive or negative relationship?

Answer: There is a positive relationship. As study hours increase, exam marks also increase.

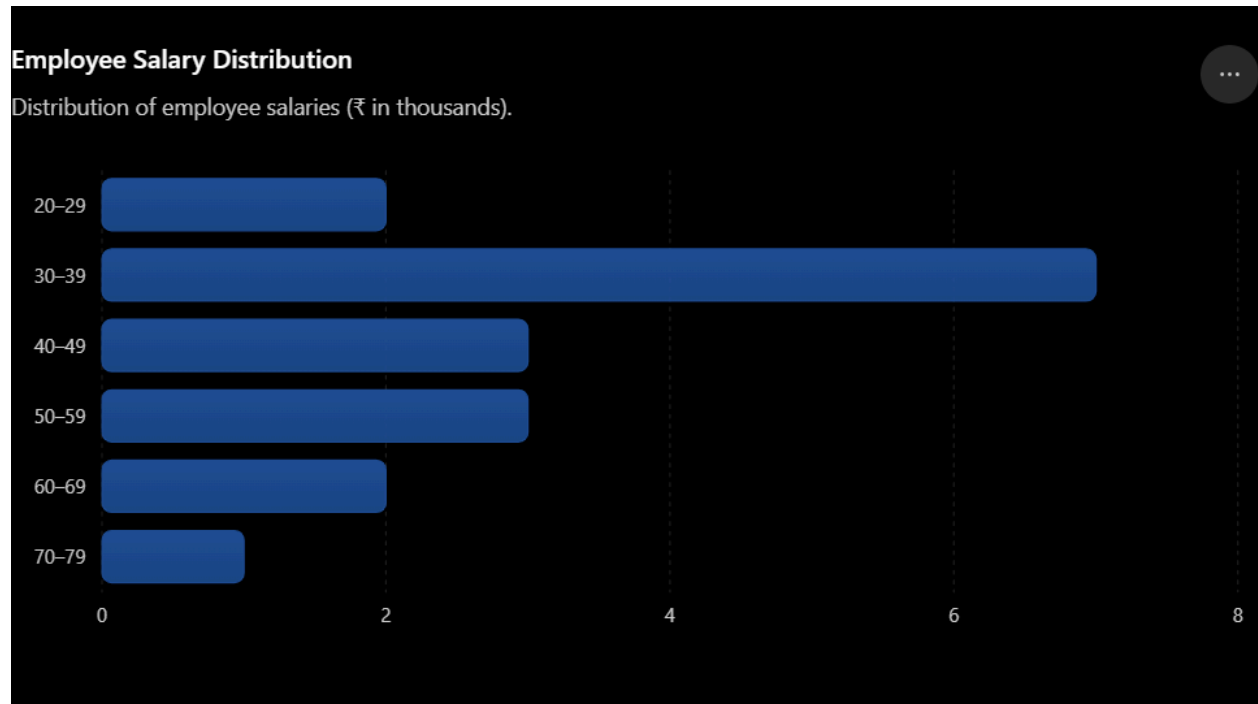
3. Which student is an outlier, if any?

Answer: There is no clear outlier. All students follow the increasing trend.

4. Can marks be predicted from study hours?

Answer: Yes. Since there is a positive relationship, exam marks can be reasonably predicted based on study hours.

### 3.Case Study: Employee Salary Distribution



1. Which graph should be used?

Answer: Histogram (or Bar Chart) is the most suitable graph to show the salary distribution.

2. In which salary range do most employees fall?

Answer: ₹30,000 – ₹39,000, with 7 employees.

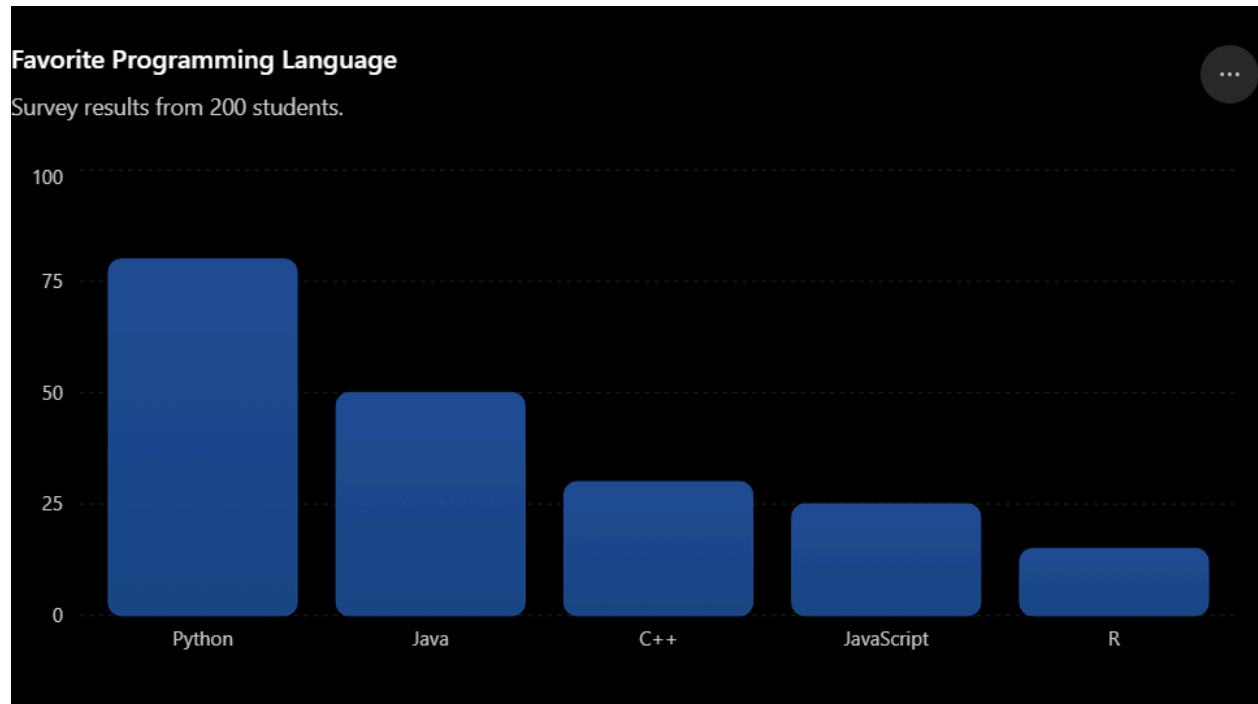
3. Is the distribution symmetric or skewed?

Answer: The distribution is slightly right-skewed (positively skewed) because there are fewer employees in the higher salary ranges (₹60,000–₹70,000).

4. Are there any salary ranges with very few employees?

Answer: Yes. The salary ranges ₹60,000–₹69,000 and ₹70,000–₹79,000 have very few employees.

#### 4. Case Study: Favorite Programming Language



1. Which visualization is suitable?

Answer: Bar Chart, because it compares the number of students who prefer each programming language.

2. Which language is most popular?

Answer: Python is the most popular language with 80 students.

3. Which language is least preferred?

Answer: R is the least preferred language with 15 students.

4. Compare Python and Java popularity.

Answer: Python is more popular than Java.

- Python: 80 students
- Java: 50 students
- Difference: 30 students more prefer Python than Java.

## 5. Case Study: Household Monthly Expenses

## Household Monthly Expenses

Percentage distribution of monthly household expenses.



● Education ● Entertainment ● Food ● Rent ● Transportation

1. Which chart best represents the data?

Answer: Pie Chart, because it shows how each expense category contributes to the total budget.

2. Which category consumes the largest budget?

Answer: Rent consumes the largest budget with 40%.

3. What percentage is spent on Food and Education together?

Answer:

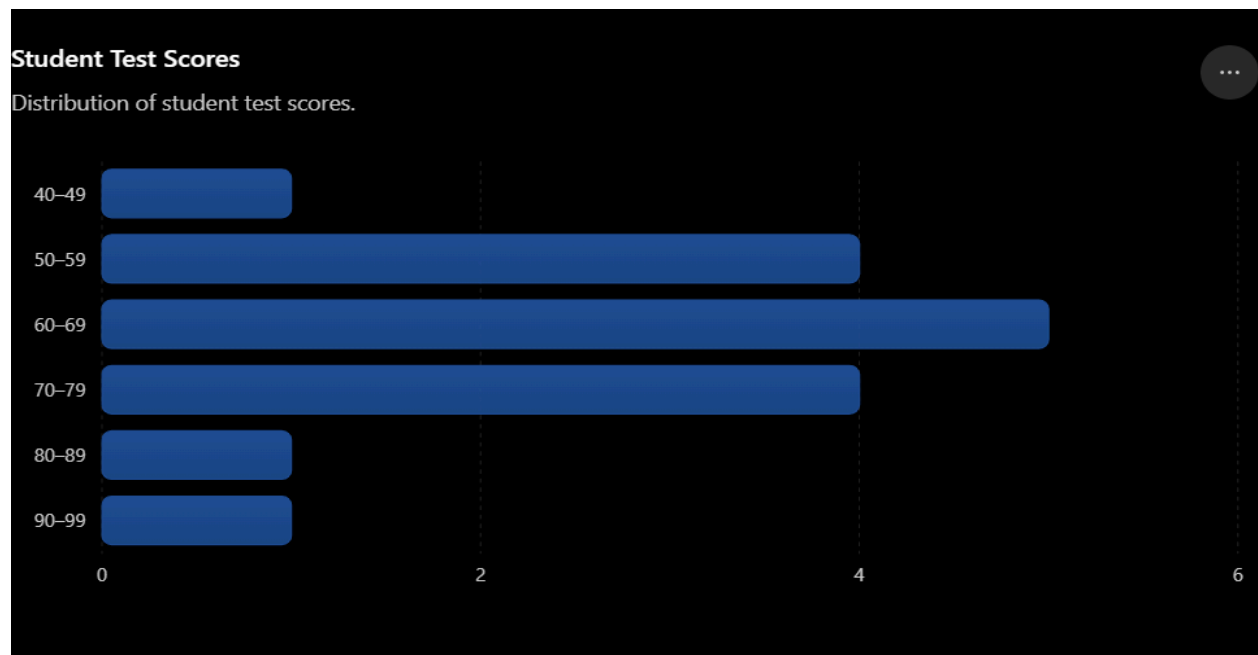
- Food = 25%
- Education = 15%

Total = 40%

4. Which categories occupy the smallest share?

Answer: Transportation and Entertainment each occupy the smallest share with 10% each.

## 6.Case Study: Student Test Scores



1. Which plot should be used?

Answer: Box Plot is the most suitable plot because it shows the median, quartiles, spread, and outliers.

2. What is the median score?

Answer: There are 15 scores, so the median is the 8th value.

Median = 65

3. Are there any outliers?

Answer: Yes. The score 95 is a possible outlier because it is much higher than the other scores.

4. What is the spread of the data?

Answer:

- Minimum score: 45
- Maximum score: 95
- Range =  $95 - 45 = 50$

Spread of the data = 50 marks.